Webscraping Project

University of Lucerne

Faculty of Humanities and Social Sciences

Institute of Computational Social Sciences

Supervised by:

Dr. Andrea de Angelis

Webscraping with R (FS 26)

Submitted by:

Reachel Klamt

my Streetname x, 6005 Lucerne

+41 xx xxx xx xx

reachel.klamt@stud.unilu.ch

Matrikelnummer: 21-118-153

Place and Date:

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Abstract

This paper examines whether Investor-State Dispute Settlements (ISDS) reproduce the asymmetric power dynamics described by dependency theory. The dependency theory claims that in a capitalist and more specifally, a neoliberal system, multinational corporations are the primary actors extracting value from developing economies to earn profit. By using data that I webscaped from the International Centre for Settlement of Investment Disputes this paper tests whether ICSID arbitration is disproportionately used by investors from high-income states against developing country respondents. The results show that the overall use of the ICSID cases do not confirm the theory. However, when only looking at extractive sectors, high-income claimants disproportionately sue upper-middle-income respondents. This provides partial support for the dependency theory hypothesis.

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1 Introduction

In this paper I argue that Investor-State Dispute Settlements are an institution which further deepens the asymmetric power dynamics inherent in the international economic order. These ISDS allow investors (mostly private companies) to sue states for alleged violations of investment agreements. On the one hand this is clearly a threat to local autonomy and democratic self-determination, because it gives companies the power to go against any policies that would potentially decrease their profit. There have been various cases where companies sued for their lost profit because these states implemented "green policies" demanded by its citizens. While this in itself is deeply problematic, the focus of this paper is on how ISDS's could potentially hinder the development of developing countries by reinforcing existing global economic hierarchies.

This paper builds upon dependency theory as its theoretical framework. The idea of the dependency theory is that the foreign capital that comes from developed capitalist centers, mainly western states, creates an imbalance between them and their periphery (developing states). The profits that are produced in the periphery are appropriated by the capitalist center. This makes the creation of an independent economy in the developing states a hard task, while the developed regions can further strengthen their economic dominance (Frank 1967; Ghosh 2019; Tausch 2010). In his study, Tausch (2010) finds that in the long term, higher multinational capital penetration leads to greater inequality and lower performance on standard indicators of development as well as lower predicted economic growth. These findings provide some empirical evidence for the dependency theory. By linking the idea of the dependency theory to more recent works on neoliberalism such as Harvey (2019) and Slobodian (2018) , I argue that these extractive powers no longer come from the economically stronger western states themselves but are increasingly exercised by from companies that exist within these developed countries. ISDS serve as the institutional infrastructure through which these corporations can enforce their economic interests against sovereign states

If this theory holds, I would expect ISDS to be used disproportionately by investors from high-income states against developing country respondents. This paper tests this hypothesis empirically using data from the International Center for Settlement of Investment Disputes (ICSID).

2 Method

2.1 Data Collection

In order to collect the different ISDS cases, I combined a API request with a web scraping approach.

The first step in the data collection process was to collect all case ID's, since they would be later needed to loop through all the case links. Fortunately it was farly easy to do so through a simple undocumented (internal) API request, because although the ICSID does not have a documented API, there webpage is built by calling a internal API. To collect the data, I simply created a GET request to the same API, which returned structured JSON data on all registered cases.

```
# make a GET request to the API endpoint
  http_response <- request("https://icsid.worldbank.org/api/all/cases")
 ↪  |>
    req_perform()
```

I made the request and unpacked the response using the `httr2` package.

Since not all the information needed for this analysis were contained in this initial dataset, the second step was to create a web-scraper that extracted the information from every single case internet page. As mentioned above, the web links to these sites consisted of the base-url and the case ID so looping through them was fairly easy. I first created a vector with all the case IDs and then a function that put together the weblink and then went to the website and scraped the required information.

```
# defining the base url
base_url <-
 ↪  "https://icsid.worldbank.org/cases/case-database/case-detail?CaseNo="



# defining a function to scrape case details
  scrape_detail <- function(case_no) {
    Sys.sleep(0.2)  # wait 0.2 seconds after every request to avoid
     ↪  overwhelming the server
```

```
  url  <- paste0(base_url,case_no) # construct the url for the case
↪ details page using the case number

  # tryCatch catches the error before it crashes the loop
  page <- tryCatch(
    read_html(url),
    error = function(e) {
      message("Failed: ", case_no, "  ", e$message)
      return(NULL)
    }
```

The first couple of times I ran the function, it failed and then stopped so I created a second part (not displayed above) which would catch errors before the function would crash and just fill everything in NA.

Within this function, there is a second function (`get_field`) which parses the html code for the required information and builds a single row out of this information. When running the function for all the case IDs

```
case_details <- map_dfr(case_nos, scrape_detail, .progress = TRUE)
```

the `map_dfr` directly puts the row from each case ID directly into a data frame, so that I don't end up with hundreds of single tibbles that each only contains one row. With that the data collection process is already completed.

2.2 Data Cleaning

The first step of the data cleaning was to make one dataset out of these two raw data collections. To do so, I basically had to extract the information from the JSON file and combine it with the data frame from the web scraping. I did so by creating a function:

```
cases_df <- map_df(cases_list, function(case) {

  # extract the first element of the caseproceedings list, which contains
  ↪ the date_registered information
```

```
  proc <- case$caseproceedings[[1]]

  tibble(
    case_no         = case$caseno,
    claimant        = case$claimant,
    respondent      = case$respondent,
    date_registered = as.Date(mdy_hms(proc$dateregistered)),
    status          = case$status
  )
})
```

The function is similar to the one I used for the web scraping. I first looked where the important information were stored and then created a tibble for each case that contained all these extracted information. It was then again mapped to a data frame.

In a second step I had to do some data cleaning and because for my analysis I needed the countries to be stored in a standardized format. For the respondent countries this was a straightforward task, since almost all of the respondents were states and were stored as such. For the few exceptions where the respondents were not a state, a replaced the value with NA because these cases are not of interest. To convert the states to iso3c standardized format I used the r-package `countrycode`, which takes different types of input (in this case `state.name`) and gives back a defined output (in this case `iso3c`). There were only very little exceptions for which I had to match the state to the iso3 format manually.

For the claimants this task was substantially harder, because only the nationality of the sueing company was written down in brackets. One one hand this made the `countrycode` packages useless and it furthermore required the extraction of the nationality from the entire variable field.

Since the nationality was in brackets, I could just use a regular expression to extract it. The solution for the second problem required the use of another database, which I found on Github. This table stored each state and how the nationality was called and even the iso3c notation. So downloaded that table and wrote a function that matched the nationality from my database with the nationality of the downloaded github table and extracted the corresponding iso3c notation. However since not all the nationalities in my dataset were stored as nationalities but some as countries, I created a fallback, that if in case the nationality did not match, it should check the

country name.

```
# Try to match claimant company to a country
icsid_df <- icsid_df |>
  mutate(
    iso3_claimant = coalesce(
      # Step 1: match against nationality adjectives (e.g. "Dutch",
       ↪ "French")
      nationality_lookup$iso3[match(claimant_nationality,
↪ nationality_lookup$Nationality)],
      # Step 2: fall back to country name match (e.g. "Luxembourg",
       ↪ "Seychelles")
      nationality_lookup$iso3[match(claimant_nationality,
↪ nationality_lookup$Name)]
    )
  )
```

Unfortunately, this method was not as reliable as the one that worked with the respondents, so I needed to match various cases manually. A second problem that occurred, was that some cases had more than one claimant. While this is probably not the statistically best solution, I ended up creating multiple rows, for each claimant one, for the cases with two or more claimants.

Furthermore, since I was not sure if I wanted to do some grouping by region later on in the analysis, I decided to already integrate for the World Bank region and UN subregion for both the claimant and respondent country into the dataset during the data cleaning process. This was a easy process, since the `countrycode` package contains these regions.

Last but not least, I needed to classify countries according to their economic status. Several approaches exist for this purpose, such as GDP per capita, the Human Development Index, or simple continent-based groupings. I decided to use the World Bank income classification because it is a widely recognized and standardized measure used extensively in international economics and development research, making it particularly well suited for testing hypotheses derived from dependency theory. It classifies countries into four categories based on Gross National Income (GNI) per capita: Low income, Lower middle income, Upper middle income, and

High income. These four categories map naturally onto the core-periphery framework of dependency theory, with High income countries representing the core and Low and Lower middle income countries representing the periphery. GNI is more appropriate than GDP for measuring a country's wealth because it captures income flows across borders, which is particularly relevant in the context of dependency theory where profit extraction from the periphery to the core is a central concept.

The `wbstats` package directly uses the World Banks API, so using this package instead of another API request saved me a lot of work and made the process less complicated.

```
# Load World Bank income category data
income_df <- wb_countries() |>
  select(iso3c, income_level)

# add the columns to the df
icsid_df <- icsid_df |>
  left_join(income_df |> rename(income_claimant  = income_level),
            by = c("iso3_claimant"  = "iso3c")) |>
  left_join(income_df |> rename(income_respondent = income_level),
            by = c("iso3_respondent" = "iso3c"))
```

First, a additional data frame, that contained the iso3c and the income classification from the world bank was created, using the `wbstats` package. By matching the `iso3_claimant` and `iso3_respondent` variables in my ICSID dataset against the ISO3 codes in the World Bank income data frame looking up the corresponding income group, the two new variables were created.

3 Analysis

As described above, to analyze the data I use the ISCID cases combined with a categorization of the income group by the world bank.
In the first step, the overall distribution of ICSID cases across claimant and respondent income groups is examined. In the second step, the analysis is narrowed to extractive sectors, where the dependency theory argument is theoretically most plausible.

To test whether dispute patterns deviate from what would be expected if they would be used

at random and not in any pattern, a chi-square test of independence is used. The chi-square test is appropriate because both the claimant and respondent income group are categorical variables and the hypothesis tests whether the distribution is random or follows a pattern. However, a significant chi-square result only tells us that an association exists but it does not tell where in the table that association comes from. To address this, I use standardized residuals. A standardized residual greater than +2 indicates that a combination occurs significantly more often than expected, while a value below -2 indicates significant under-representation.

In a second step I classify the different sectors into extractive sectors and non-extractive sectors with the first containing sectors such as oil, water, agriculture and forestry because these sectors are characterized by the exploitation of natural resources that are geographically fixed making them particularly prone to disputes between foreign investors and host states. The latter group contains disputes regarding the financial sector, information and communication sector and the energy sector.By isolating extractive cases, it is possible to test a more refined version of the dependency theory hypothesis: even if high-income claimants do not disproportionately target poorer states overall, they may do so specifically in sectors where natural resource extraction is at stake. The chi-square test and standardized residuals are applied to this subset in the same way as in the first step.

To add to the analysis of the income groups, I also examine how the use of ICSIDs have developed over tiem. The cases are categorized into three dispute patterns that are based on the income group of the claimant and respondent. I grouped high-income claimants versus developing respondents (low, lower middle, and upper middle income), high-income claimants against high-income respondents, and all other combinations. A total case count is included as a reference. To smooth out year-to-year variation and make long-run trends more visible, LOESS regression lines are fitted to each pattern alongside the raw data. The LOESS-regression functions similar to a normal regression but it allows for non-linearity as well.

4 Results

4.1 Claimant vs. Respondent Countries

To get a overview of how the cases are distributed among the world, I first plotted the number of how many cases each country was involved in, an ICSID case as a claimant and as a respondent. In Figure 1. The North-South divide is clearly visible, with European countries and the US clearly being more involved in this disputes as claimants and developing countries, especially

in Latin America displaying a much larger number in the respondent position. However, based on this plot, we cannot see if countries tend to be sued by investors from similar or different regions because the maps only show the total number of cases per country.

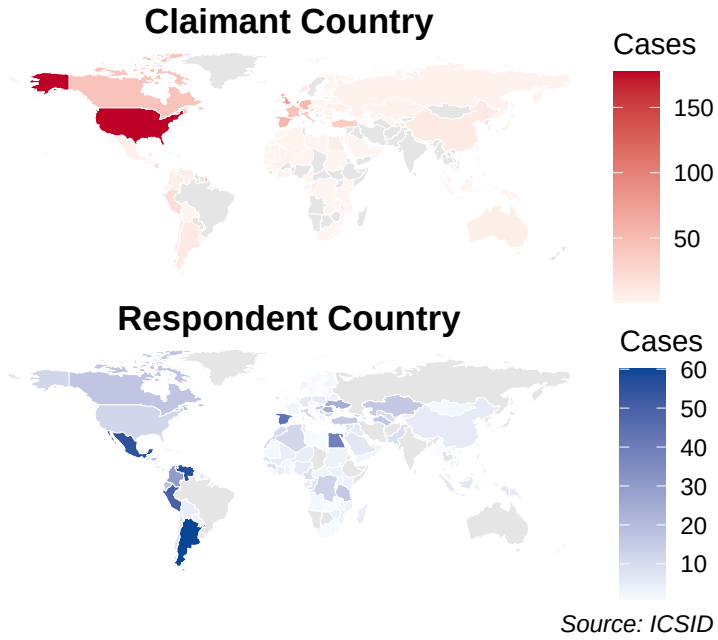
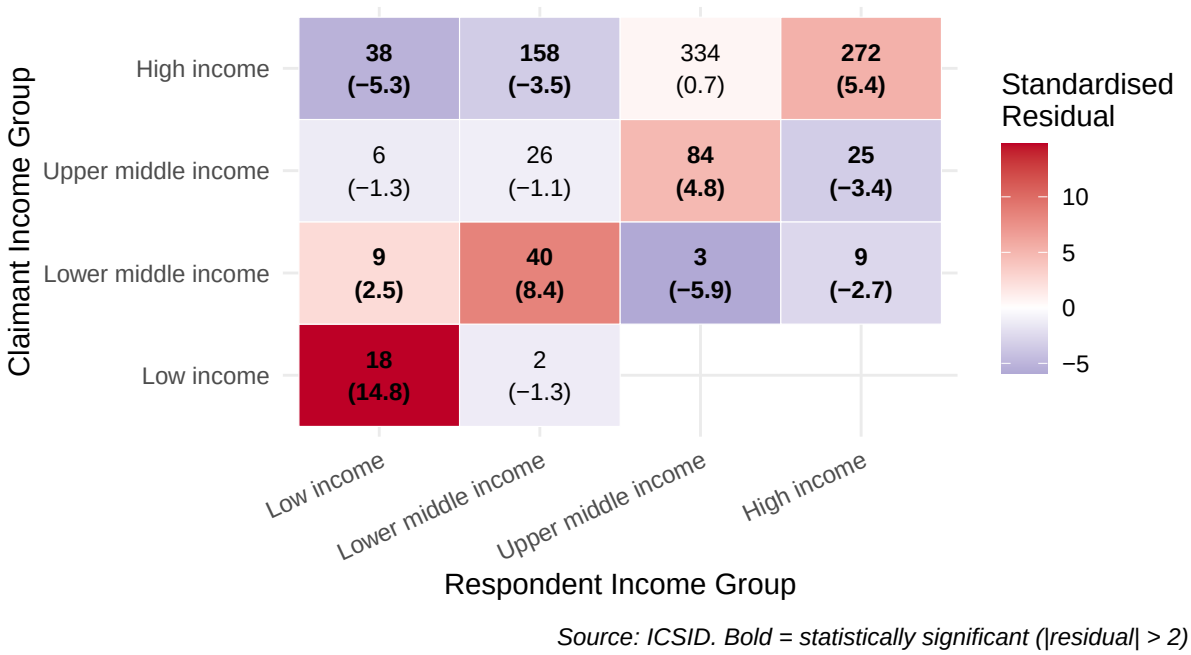


Figure 1: ICSID Cases by Claimant and Respondent Country

4.2 Claimant vs. Respondent by Income Group

So the next step, to investigate the hypothesis and to quantify these result, was to check if there is a correlation between the number of cases the different income groups that were suing each other. As mentioned above, to do so, I used the income classification for each country from the world bank. Not included in this report, there is a plot, which shows the income classification for each country.

I used a chi-squared test to see if there is any correlation. With a p-value of 2.2e-16 the test shows that there is a significant correlation between the income group of the claimant and the income group of the respondent. This means that there is a pattern and that it is not random whom sues who. In Figure 2 these patterns are made visible.



Source: ICSID. Bold = statistically significant (|residual| > 2)

Figure 2: ICSID Cases by Income Group

When looking at the number of cases, one can see that the ICSD is a tool that mainly high- and upper middle income countries use. This makes sense because these are the countries that host the majority of multinational corporations with the financial resources that are required to pursue international arbitration. Moreover, a international dispute settlement is a expensive process, so companies seated in lower-income countries just cannot afford these proceedings.

Second, when looking at the standardized residuals from the chi-square test we find that disputes among claimants and respondents of the same income class are over-represented while high-income claimants suing lower-income respondents are either neutral or negative. This suggests that the expected pattern of wealthy states suing developing states cannot be found.

4.3 Claimant vs. Respondent by Income for Extractive Sectors

Figure 3 shows that, compared to the Figure 2, in extracitve sectors high income claimants do disproportionately sue upper middle income countries. One caveat of this analysis that has to be mentioned, is, that the sample size is getting smaller and therefor statistical power decreases.

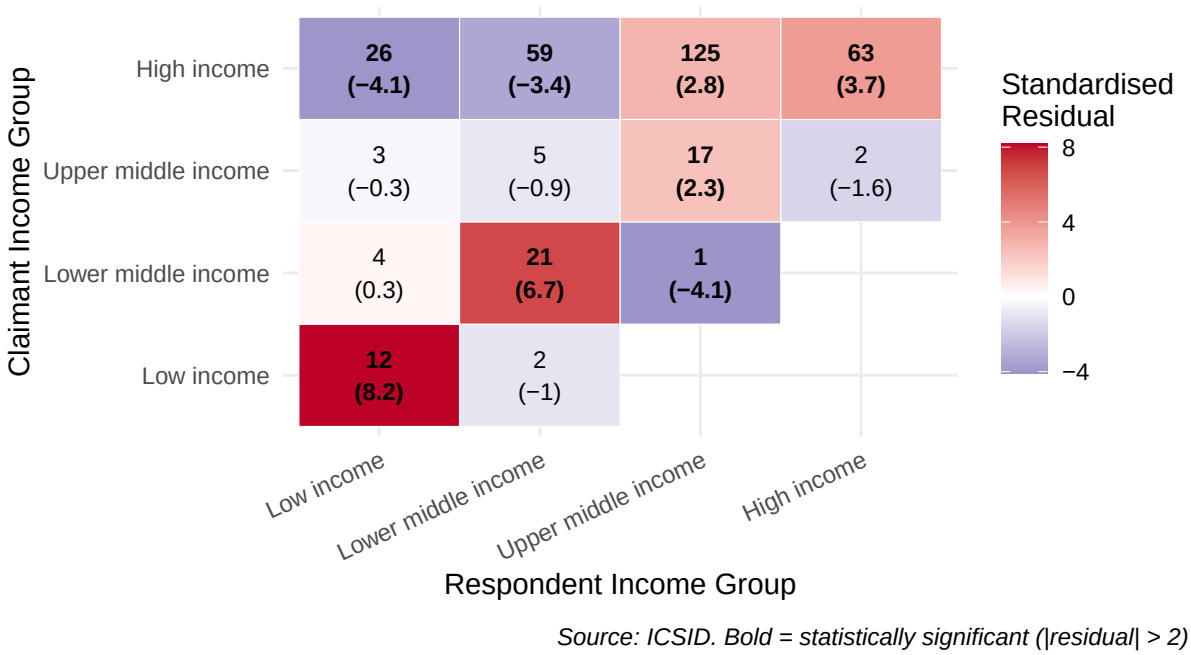


Figure 3: ICSID Cases by Income in extractive sectors

The heatmap shows that in the extractive sector we find partial support for the argument that ICSID cases are a tool used by multinational corporations. However, the effect is concentrated against upper-middle-income states not the poorest countries. This suggests that investment arbitration in the extractive sector functions less as a mechanism of exploitation in the classical sense, and more as a tool for wealthy investors to challenge state sovereignty over natural resources.

4.4 Time Trend

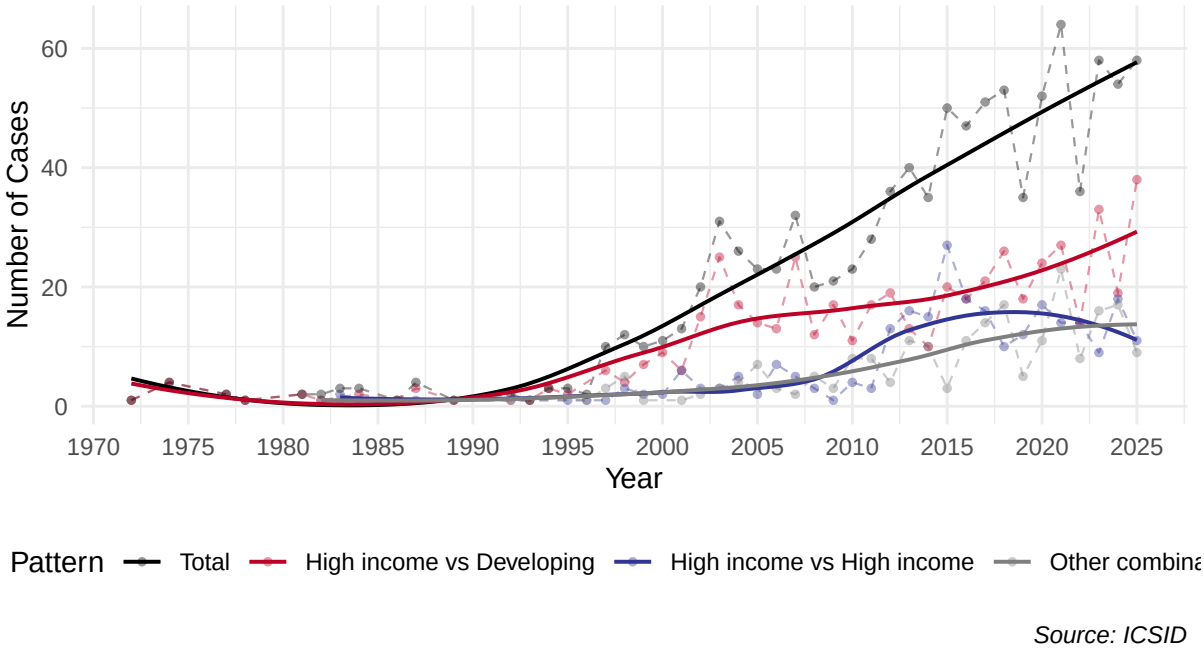


Figure 4: ICSID Cases by Sector and Income

The dominant pattern throughout the entire period is that high-income claimants are suing mostly developing respondents. However, high-income against high-income disputes have grown since 2000. Disputes where the claimant is not from a high-income country remain low throughout the entire period. This is consistent with the earlier finding that lower-income claimants rarely use ICSID arbitration. Overall, It is also notable that the numbers of cases per year have been increasing since the 1990s.

5 Conclusion

While the findings of the analysis that included all sectors did not clearly support the hypothesis of this paper, when only looking at the extractive sector the paper finds patterns that do provide partial empirical support for the dependency theory argument. Specifically, high-income claimants disproportionately sue upper-middle-income respondents in extractive sectors. The dominant statistical finding across all sectors is that investors tend to be sued investors from similar income groups.

The effect, where it exists, is concentrated against upper-middle-income states such as Argentina, Venezuela, and Ecuador, rather than the least developed countries.

Several limitations of this analysis should be acknowledged. First, the data is limited to ICSID cases only and does not capture disputes filed under other arbitration forums such as

UNCITRAL, ICC, or the Stockholm Chamber of Commerce, which may exhibit different patterns. Second, the income classifications used are based on current World Bank data and may not accurately reflect the economic status of countries at the time the disputes were filed. Future research could address these limitations by expanding the dataset to include other arbitration forums and by incorporating time-varying income classifications to better capture the economic context at the time of each dispute.

Important note: Non of the data was pushed to Github. The datasets were excluded by putting the data folders (`data_raw` and `data_preprocessed` into the .gitignore
Additional note: I also did not push my Latex-Template File because it contains personal information which I do not want to be visible to everybody. Moreover I did not push the References.bib and the apsa.csl, the latter because it is not my work and I don't know if I would even be allowed to that and the first one because it is updated automatically by zotero, so if I add literature, even if I don't use it in that report, would need a new commit.)

6 References

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7 Declaration of AI usage

In the process of creating this paper the LLM model Claude (Sonnet 4.6) was used for several tasks.

1. **Generating and improving my R Code**

   I used LLM to help me generate the webscraping loop and write the debug fixes after the initial crashes. In the data cleaning process i used an iterative approach of writing code, giving its output to LLM and based on its output improving the code so that with each iteration, I got closer to the clean data frame that I aimed for. Lastly, I relied on the LLM to generate the base code for the visualizations I wanted to create, which I then adjusted to fit my style (for example adjusting the position and content of the caption). To do so I specifically ask the LLM

2. **Summarizing technical documentation and research papers**

   I used the LLM to help me in the process on deciding which would be the best statistical approach for analyzing the data other than the normal chi-squared test. I confirmed the answers (standardized residuals) by reading on websites that explain what standardized residuals are.

3. **Exploring available tools**

   When looking for a way to classify countries by income groups, I asked the LLM for different approaches. One of the recommendation was the wbstat-package which I've already heard about and considered it to be a good fit.

4. **Checking ethical and legal aspects of the work**

   I used the LLM to help me interpret the robots.txt from different websites I considered to scrape.

5. **Improving language**

   I used the LLM to improve sentences or finish a sentence I began, when I could not find good expressions or lacked some words on what I wanted to say. When I used the LLM in this way, I would always enter my idea or the beginning of the sentence with the prompt to make it sound more scientific or to find a expression the replace the "…"

8 Declaration of Originality

"I hereby declare that I have written this thesis independently and have not used any sources other than those indicated. I have marked as such all passages taken verbatim or in spirit from sources."

Luzern, 10.04.2026

Place and Date

Signature